

Question Number	Answer	Additional guidance	Mark
5(a)(i)	• Tabebuia		(1)

Question Number	Answer	Mark
5(a)(ii)	The only correct answer is A amyloplast <i>B is incorrect because middle lamella is not a structure that stores starch</i> <i>C is incorrect because plasmodesmata is not a structure that stores starch</i> <i>D is incorrect because tonoplast is not a structure that stores starch</i>	(1)

Question Number	Answer	Additional guidance	Mark
5(b)(i)	An answer that includes the following points: • slime capsule prevents dehydration of cell (1) • pili allow bacteria to adhere (to surfaces) (1)	Accept protection of cell from a correct factor e.g. {white blood cells / phagocytes / antibodies / antibiotics / from harsh conditions} Accept conjugation / exchange of genetic information / exchange plasmids / attachment Ignore sexual reproduction	(2)

Question Number	Answer
5 (b)(ii)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • compound needs to be extracted from trees and used to make a drug • compounds tested on <i>Y. pestis in vitro</i> • tested on {animals / human cells} • (phase 1 / preliminary) (small scale) tests on healthy {people / volunteers} • review by independent {scientists / medics} to see if work can progress to stage 2 • (phase 2) - drug tested on {small / 100 to 500} groups of {patients / people} who have the {plague / disease} • appropriate concentrations identified • (phase 3) - drug tested on {larger groups of / 1000 to 3000} {patients / people} who have the plague • placed randomly in two groups - one group receives {treatment / drug containing the chemical compounds} and the other receives placebo • double blind test • analyse results with (appropriate) statistical test / test for significant difference

			(6)
			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Demonstrates isolated elements of biological knowledge related to the given context with generalised comments made. The description will contain basic information with some attempt made to link knowledge and understanding to the given context.	outline of drug test given but no reference to either the plague or <i>Y. pestis</i> or bacteria core prac only = 1 mark other human drug trial = 2 marks
Level 2	3-4	Demonstrates adequate knowledge by selecting and applying some relevant biological facts/concepts to provide the description being presented. The description shows some linkages and lines of scientific reasoning with some structure.	Testing on {animal / human cells} plus at least two other phases correctly described In context of people having the disease / plague or <i>Y. pestis</i> or bacteria gives lower mark plus reference to double blind test or placebos gives higher mark
Level 3	5-6	Demonstrates comprehensive knowledge by selecting and applying relevant knowledge of biological facts/concepts to provide the description being presented. The description is clear, coherent and logically structured.	Testing on {animal / human cells} plus three phases correctly described including double blind test and placebos In context of people having the disease / plague or <i>Y. pestis</i> or bacteria gives lower mark plus correct ref to use of stats / analysis of significant difference to placebo gives higher mark

			or in vitro testing of drug on <i>Y. pestis</i> or bacteria gives higher mark
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